Def. Let V be a vector space over $\mathbb{R}$ (or $\mathbb{C}$).

An <u>inner product</u> on V is a function $\langle \cdot, \cdot \rangle$ which takes pairs of vectors in V to scalars in $\mathbb{R}$ (or $\mathbb{C}$). In addition the function satisfies

1. $\langle u, v \rangle = \overline{\langle v, u \rangle}$ $\quad \forall u, v \in V$.

2. $\langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ $\quad \forall u, v, w \in V$.

3. $\langle cu, v \rangle = c \langle u, v \rangle$ $\quad \forall c \in \mathbb{R}$ (or $\mathbb{C}$), $u, v \in V$.

4. $\langle u, u \rangle \geq 0$ and $\langle u, u \rangle = 0 \iff u = 0$ (zero vector)

A vector space with an inner product is called an <u>inner product space</u>.

Example 2 in textbook

Take $V = \mathbb{P}_n$ = polynomials of degree n or less with real coefficients

Pick $n+1$ distinct real numbers $t_0, \dots, t_n \in \mathbb{R}$.

Define $\langle p, q \rangle = p(t_0)q(t_0) + \dots + p(t_n)q(t_n)$ $\quad (*)$

for any $p, q \in \mathbb{P}_n$.

Check properties 1-4 in def'n of inner product space. For property 4, observe that

$$0 = \langle p, p \rangle \iff 0 = (p(t_0))^2 + \cdots + (p(t_n))^2$$

$$\iff p(t_0) = \cdots = p(t_n) = 0.$$

$\Rightarrow$ p has $n+1$ distinct roots (zeros) $t_0, \ldots, t_n$.

$\Rightarrow$ $p = 0$ since $\deg p \leq n$.

Fact: A nonzero polynomial of degree n can have at most n distinct roots.

We conclude that $\langle \cdot, \cdot \rangle$ as defined in (*) above is an inner product on $\mathbb{P}_n$.

Example 3 in textbook Fix $t_0 = 0, t_1 = 1/2, t_2 = 1$.

Take $n = 2$ and $V = \mathbb{P}_2$ = polynomials of degree 2 or less.

Take $p(x) = 12x^2$, and $q(x) = 2x - 1$. Let's compute $\langle p, q \rangle$ as in (*).

$$\langle p, q \rangle = p(0)q(0) + p(1/2)q(1/2) + p(1)q(1).$$

$$= (12(0))(2(0)-1) + 12(1/2)^2(2(1/2)-1) + 12(1)(2(1)-1)$$

$$= 0 + 0 + 12$$

$$= 12$$

$$\langle q, q \rangle = (q(0))^2 + (q(1/2))^2 + (q(1))^2$$

$$= (-1)^2 + 0^2 + 1^2$$

$= 2$

Example 7 in text book.

Take $V = C[a,b]$ = the continuous functions defined on the interval $[a,b] \subseteq \mathbb{R}$.

Define $\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt, \qquad f, g \in C[0,1]$

Let's check property 1 of the def'n of inner product.

$$\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt = \int_a^b g(t)\, f(t)\, dt = \langle g, f \rangle$$

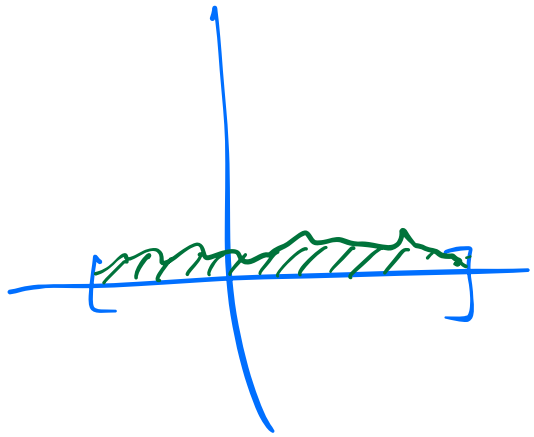
Let's check property 2.

$$\begin{aligned}
\langle f+g, h \rangle &= \int_a^b (f+g)(t)\, h(t)\, dt \\
&= \int_a^b (f(t) + g(t))\, h(t)\, dt \\
&= \int_a^b f(t)\, h(t) + g(t)\, h(t)\, dt \\
&= \int_a^b f(t)\, h(t)\, dt + \int_a^b g(t)\, h(t)\, dt \\
&= \langle f, h \rangle + \langle g, h \rangle .
\end{aligned}$$

For property 4., let's consider

$$\langle f, f \rangle = \int_a^b f(t)\, f(t)\, dt$$

$$= \int_a^b (f(t))^2\, dt.$$



= area under the nonnegative function f^2 on $[a,b]$

≥ 0.

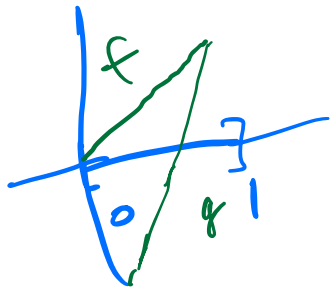
Moreover $\langle f, f \rangle = 0 \iff$ the area under f^2 is zero

$\iff f^2(t) = 0$ (zero function) $\forall t \in [a,b]$

$\iff f(t) = 0 \quad \forall t \in [a,b]$.

Example 8 in textbook.

Take $a = 0$ and $b = 1$, $V = C[0,1]$



Take $f(x) = x$ and $g(x) = 2x - 1$

$\forall x \in [0,1]$

$$\langle f, g \rangle = \int_0^1 t(2t-1)\, dt$$

$$= \int_0^1 2t^2 - t\, dt$$

$$= \left[\frac{2}{3} t^3 - \frac{t^2}{2} \right]_0^1$$

$$= (2/3 - 1/2) - (0 - 0) = \frac{4}{6} - 3/6 = 1/6.$$

f is not orthogonal to g since $\langle f, g\rangle \neq 0$.

$$\langle f, f\rangle = \int_0^1 t^2\,dt = \left[t^3/3 \right]_0^1 = 1/3 .$$

Remark - The main idea for these new inner product spaces is that everything we did for the standard inner products on $\mathbb{R}^n$ and $\mathbb{C}^n$, still work!

In particular we have norm, projection, Gram Schmidt etc.

Example

Take $V = C[-1,1]$, $f(x) = x \quad \forall x \in [-1,1]$

Then $\|f\|^2 = \langle f, f\rangle = \int_{-1}^1 t^2\,dt = \left[t^3/3 \right]_{-1}^1 = \frac{1}{3} - \frac{(-1)^3}{3} = 2/3$

$\Rightarrow \|f\| = \sqrt{2/3}$

Example Gram Schmidt (Exercise 31 §6.7)

Take $V = C[-1,1]$, $p_0(x) = 1$, $p_1(x) = x$, $p_2(x) = x^2$

$\forall x \in [-1,1]$.

Let $u_0 = p_0$. Let $u_1 = p_1 - \frac{\langle p_1, u_0\rangle}{\langle u_0, u_0\rangle} u_0$

$$= p_1 - \frac{\int_{-1}^{1} t\,dt}{\int_{-1}^{1} dt} u_0$$

$$= p_1 - \frac{\left[t^2/2\right]_{-1}^{1}}{\left[t\right]_{-1}^{1}} u_0$$

$$= p_1 - \frac{0}{2} u_0$$

$$= p_1 .$$

Let $u_2 = p_2 - \frac{\langle p_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle p_2, u_0 \rangle}{\langle u_0, u_0 \rangle} u_0$

$$= p_2 - \frac{\int_{-1}^{1} t^2 (t)\,dt}{\int_{-1}^{1} t^2\,dt} u_1 - \frac{\int_{-1}^{1} t^2\,dt}{\int_{-1}^{1} dt} u_0$$

$$= p_2 - \frac{\left[t^4/4\right]_{-1}^{1}}{\left[t^3/3\right]_{-1}^{1}} u_1 - \frac{\left[t^3/3\right]_{-1}^{1}}{\left[t\right]_{-1}^{1}} u_0$$

$$= p_0 - \frac{0}{2/3} u_1 - \frac{2/3}{2} u_0$$

$$= p_0 - 1/3\, u_0$$

$\Rightarrow \quad p_2(x) = x^2 - 1/3 \quad \forall\ x \in [-1, 1].$

Conclusion: $\{u_0, u_1, u_2\}$ is an orthogonal set and a basis for $W = \text{span}\{p_0, p_1, p_2\}$.

Exercise.

Take $y(x) = x^3 \quad \forall x \in [-1,1]$.

Compute $\text{proj}_W y = \sum_{j=0}^{2} \frac{\langle y, u_j \rangle}{\langle u_j, u_j \rangle} u_j$

$\text{proj}_W y$ is a function in W. It is the best approximation of a function in W to y.

Exercise Compute the distance of y to W

i.e. $\| y - \text{proj}_W y \| = ?$